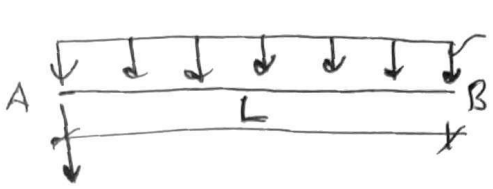
Example 8 : Determine the magnitudes and location of the resultant load.

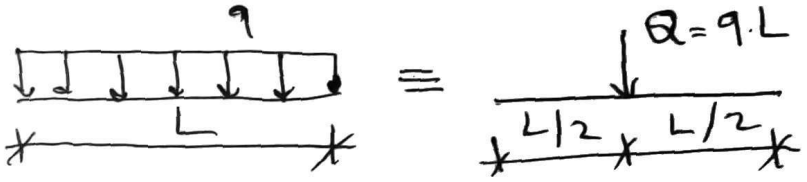


Resultand load: $Q = \int_0^L q(x)\,dx = \int_0^L q\,dx = q\int_0^L dx$

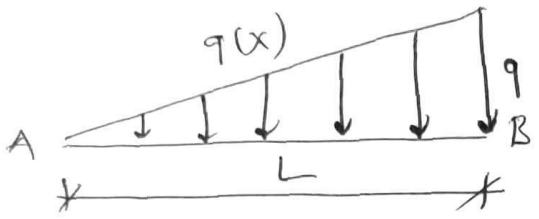
$$Q = q.x\Big|_0^L = q.L - q.0 = \boxed{q.L}$$

The location: $x_Q = \dfrac{\int_0^L q(x).x\,dx}{\int_0^L q(x).dx} = \dfrac{q\int_0^L x\,dx}{q\int_0^L dx} = \dfrac{x^2/2\Big|_0^L}{x\Big|_0^L}$

$$x_Q = \frac{\left(\frac{L^2}{2}\right)}{L} \Rightarrow \boxed{x_Q = \frac{L}{2}}$$



Example 9: Determine the magnitude and location of the resultant load.



$$q(x) = q\left(\frac{x}{L}\right)$$

The resultant load:

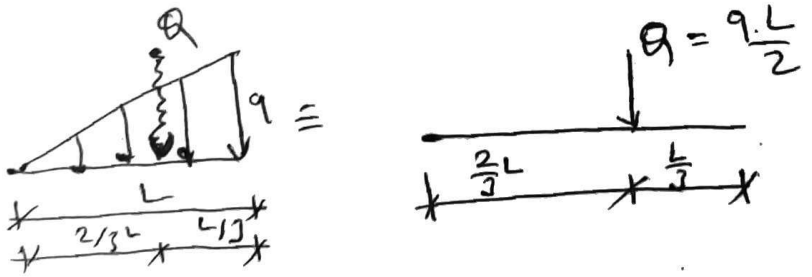
$$Q = \int_0^L q(x)\,dx = \int_0^L q\left(\frac{x}{L}\right)dx = \frac{q}{L}\int_0^L x\,dx = \frac{q}{L}\cdot\frac{x^2}{2}\Big|_0^L$$

$$\boxed{Q = \frac{qL}{2}}$$

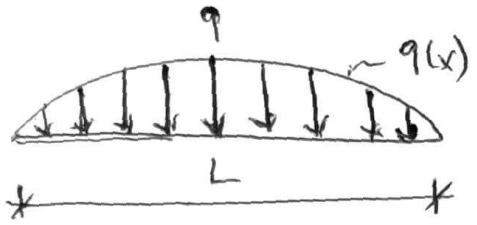
The location:

$$X_Q = \frac{\int_0^L q(x)\cdot x\,dx}{\int_0^L q(x)\,dx} = \frac{\int_0^L q\left(\frac{x}{L}\right)\cdot x\,dx}{\frac{qL}{2}} = \frac{\frac{q}{L}\int_0^L x^2\,dx}{\frac{q\cdot L}{2}}$$

$$X_Q = \frac{2}{L^2}\cdot\frac{x^3}{3}\Big|_0^L = \frac{2}{L^2}\cdot\frac{L^3}{3} \Rightarrow \boxed{X_Q = \frac{2}{3}L}$$



Example 10 : Determine the magnitude and location of the resultant load.



$$q(x) = \frac{4q}{L^2}(L-x).x$$

The resultant load:

$$Q = \int_0^L q(x).dx = \int_0^L \frac{4q}{L^2}(L-x).x.dx$$

$$Q = \frac{4q}{L^2}\left[\frac{Lx^2}{2} - \frac{x^3}{3}\right]\Big|_0^L = \frac{4q}{L^2}\left(\frac{L^3}{2} - \frac{L^3}{3}\right) = \frac{4q}{L^2}.\frac{L^3}{6}$$

$$Q = \frac{2}{3}qL$$

The location of resultant load:

Since the distribution of load is symmetrical $\rightarrow x_Q = \frac{L}{2}$

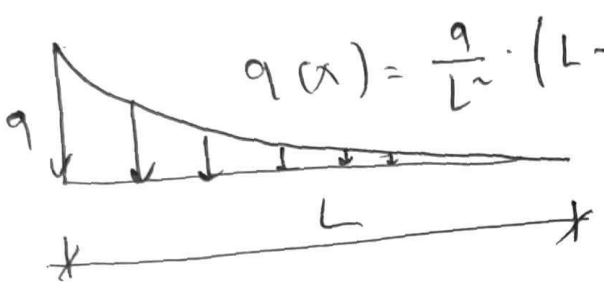
Let's prove it!

$$x_Q = \frac{\int_0^L q(x).x.dx}{\int_0^L q(x).dx} = \frac{\int_0^L \frac{4q}{L^2}.(L-x).x.x.dx}{\frac{2}{3}.q.L}$$

$$x_Q = \frac{\frac{4q}{L^2}\int_0^L (L-x).x^2.dx}{\frac{2}{3}.q.L} = \frac{6}{L^3}\left(L.\frac{x^3}{3} - \frac{x^4}{4}\right)\Big|_0^L$$

$$x_Q = \frac{6}{L^3}\left(\frac{L^4}{3} - \frac{L^4}{4}\right) = \frac{6}{L^3}.\frac{L^4}{12} = \frac{L}{2}$$

Example 11: Determine the magnitude and location of the resultant load.



The resultant load:

$$Q = \int_0^L q(x)\,dx = \int_0^L \frac{q}{L^2}\cdot(L-x)^2\,dx$$

$$Q = \frac{q}{L^2}\int_0^L \left[L^2 - 2Lx + x^2\right]dx = \frac{q}{L^2}\left[L^2 x - Lx^2 + \frac{x^3}{3}\right]\Big|_0^L$$

$$Q = \frac{q}{L^2}\cdot\left(L^3 - L^3 + \frac{L^3}{3}\right) \Rightarrow \boxed{Q = \frac{qL}{3}}$$

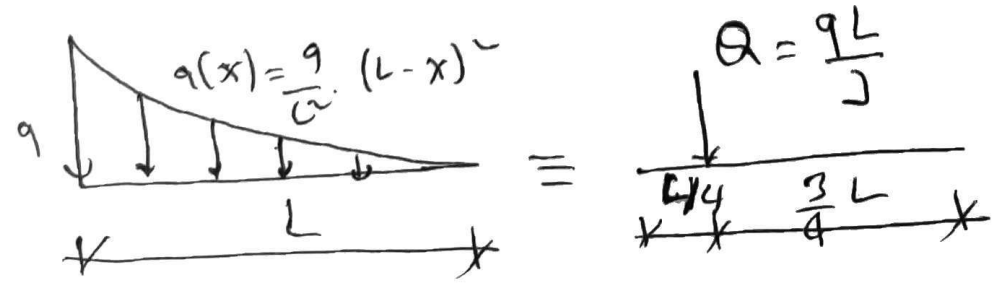
The location:

$$X_Q = \frac{\int_0^L q(x)\cdot x\,dx}{\int_0^L q(x)\,dx} = \frac{\frac{q}{L^2}\int_0^L (L-x)^2\cdot x\,dx}{\frac{qL}{3}}$$

$$X_Q = \frac{3}{L^3}\cdot\left(L^2\cdot\frac{x^2}{2} - 2L\cdot\frac{x^3}{3} + \frac{x^4}{4}\right)\Big|_0^L$$

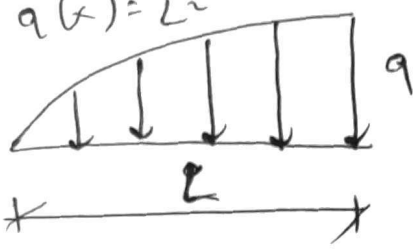
$$= \frac{3}{L^3}\cdot\left(\frac{L^4}{2} - \frac{2L^4}{3} + \frac{L^4}{4}\right) \Rightarrow X_Q = \frac{3}{L^3}\cdot\frac{L^4}{12}$$

$$\boxed{X_Q = \frac{L}{4}}$$



Example 12: Determine the magnitude and location of the resultant load.

$q(x) = \frac{q}{L^2} \cdot (2L - x) \cdot x$



The resultant load

$$Q = \int_0^L q(x)\,dx = \int_0^L \frac{q}{L^2} \cdot (2L - x) \cdot x \cdot dx$$

$$Q = \frac{q}{L^2} \left[Lx^2 - \frac{x^3}{3} \right]\Big|_0^L = \frac{q}{L^2} \left(L^3 - \frac{L^3}{3} \right) = \frac{q}{L^2} \cdot \frac{2 \cdot L^3}{3}$$

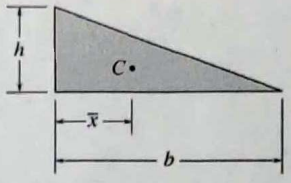
$$\boxed{Q = \frac{2}{3} qL}$$

$$X_Q = \frac{\int_0^L q(x) \cdot x\,dx}{\int_0^L q(x) \cdot dx} = \frac{\int_0^L \frac{q}{L^2} \cdot (2L - x) \cdot x \cdot x\,dx}{\frac{2}{3} \cdot q \cdot L}$$

$$X_Q = \frac{\frac{q}{L^2} \cdot \left(2L \cdot \frac{x^3}{3} - \frac{x^4}{4} \right)\Big|_0^L}{\frac{2}{3} qL} = \frac{3}{2L^3} \left(\frac{2L^4}{3} - \frac{L^4}{4} \right)$$

$$\boxed{X_Q = \frac{5}{8} L}$$

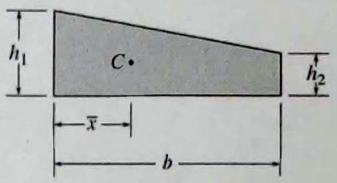
Geometric Properties of Areas



Triangle

$$A = \frac{1}{2}bh$$

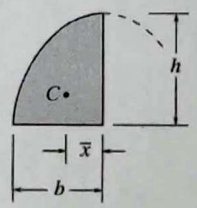
$$\bar{x} = \frac{1}{3}b$$



Trapezoid

$$A = \frac{1}{2}b\,(h_1 + h_2)$$

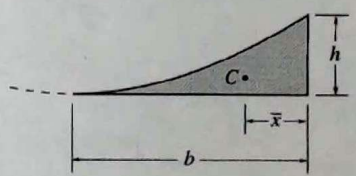
$$\bar{x} = \frac{b\,(2h_2 + h_1)}{3\,(h_1 + h_2)}$$



Semi Parabola

$$A = \frac{2}{3}bh$$

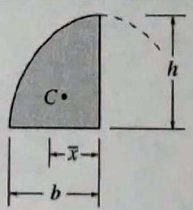
$$\bar{x} = \frac{3}{8}b$$



Parabolic spandrel

$$A = \frac{1}{3}bh$$

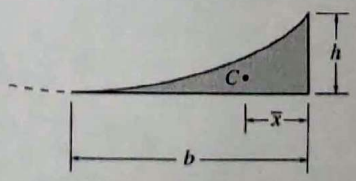
$$\bar{x} = \frac{1}{4}b$$



Semi-segment of *n*th degree curve

$$A = bh\left(\frac{n}{n+1}\right)$$

$$\bar{x} = \frac{b\,(n+1)}{2\,(n+2)}$$



Spandrel of *n*th degree curve

$$A = bh\left(\frac{1}{n+1}\right)$$

$$\bar{x} = \frac{b}{(n+2)}$$